

# Find a Real Problem

Project: Helpful Classroom Sorter

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

## 2. Practise with fading help

Use W2: Identify claims beyond the evidence for A problem worth solving. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

---

---

---

## 3. Individual reasoning check

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the outcome”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

---

---

---

---

## 4. Independent transfer case

## Independent case: Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

### My independent explanation, trace and limits

---

---

---

---

---

---

---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

## 6. Solve the given case

Shared markers are often lost because tray labels are unclear.

State the need before selecting AI.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

---

---

---

---

---

## 7. Check a changed case

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My independent answer and reason

**Write or draw your reasoning and evidence.**

---

---

---

---

---

### Week 29 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

## 9. Project checkpoint

Choose the Classroom Sorter's real user and measurable need; keep an AI-free solution option open.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

---

---

---

---

---

## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

**My independently reasoned answer**

---

---

---

---

---

---

---

Record support used: none, hint or teacher explanation.

## 12. W1 Write the problem without prescribing a tool

Use the ten-request activity log to write a problem statement with a user, setting and observed difficulty. Explain why “Build an AI app” is not yet a problem statement.

**My reasoning and evidence**

---

---

---

---

---

---

### 13. W2 Identify claims beyond the evidence

A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

**My reasoning and evidence**

---

---

---

---

---

---

### 14. W3 Choose information that serves the outcome

The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant inputs from minutes available, quiet required, pupil name and home address. State the output and a simpler alternative to a recommender. Explain one boundary.

**My reasoning and evidence**

---

---

---

---

---

---

## 15. W4 Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

### My reasoning and evidence

---

---

---

---

---

---

---

---

## 16. W5 Measure help as well as avoided errors

For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure. Name two results to record on test requests and explain why the original three-of-eight observation is not already an evaluation of the new tool.

### My reasoning and evidence

---

---

---

---

---

---

## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Find a Real Problem

A fictional class loses time finding the correct craft tray.

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

## Find a Real Problem

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Observe problem

Three mismatches are recorded out of eight requests.



#### 2. Define need

Help users choose a tray containing every requested tool.



#### 3. Set success test

Compare selected tray against the approved inventory for each new case.



#### 4. Bound the project

Use fictional tool requests; do not collect identity or purchase equipment.

### CHANGE ONE THING

The approved list contains no tray with every requested tool.

Explain the changed result using the matching lesson.

Write a one-sentence project goal and say what evidence would count as a failure.

### My picture-based explanation and changed-case prediction

---

---

---

---

---

---

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

# 19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

## Evidence to keep · My record

**Evidence to keep:** This week’s input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week’s record rather than starting again.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## 20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional class loses time finding the correct craft tray.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.

1. Bound the project
2. Observe problem
3. Set success test
4. Define need

#### My selected conclusion, deciding fact and explanation

---

---

---

---

---

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete need. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it. Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

### Practice 2: explain the deciding evidence

Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The finder should report no suitable tray rather than guess.
3. This one example guarantees correct results for every future input.

### **My selected conclusion, deciding fact and explanation**

---

---

---

---

---

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

### **Repair a missing building block**

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its output and permission to take an action. Short practice: Which original proposal is authorised? Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied evidence and checkable output requirements. Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?. Return to this week and explain what you changed.

### **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### **What I repaired, the evidence I kept and the next test**

---

---

---

---

---